

5.4.2.2—Coefficient of Thermal Expansion

The coefficient of thermal expansion should be determined by the laboratory tests on the specific mix to be used.

In the absence of more precise data, the thermal coefficient of expansion may be taken as:

- For normal weight concrete: $6.0 \times 10^{-6}/^{\circ}\text{F}$, and
- For lightweight concrete: $5.0 \times 10^{-6}/^{\circ}\text{F}$

5.4.2.3—Creep and Shrinkage

5.4.2.3.1—General

Values of creep and shrinkage, specified herein and in Articles 5.9.3.3 and 5.9.3.4, shall be used to determine the effects of creep and shrinkage on the loss of prestressing force in bridges other than segmentally constructed ones. These values in conjunction with the moment of inertia, as specified in Article 5.6.3.5.2, may be used to determine the effects of shrinkage and creep on deflections.

These provisions are applicable for design concrete compressive strengths specified in Article 5.1, including normal weight concrete up to 15.0 ksi. In the absence of more accurate data, the shrinkage coefficients may be assumed to be 0.0002 after 28 days and 0.0005 after one year of drying.

Except for segmentally constructed bridges, or where mix-specific data are not available, estimates of shrinkage and creep may be made using the provisions of any of the following:

- Articles 5.4.2.3.2 and 5.4.2.3.3,
- The *fib Model Code for Concrete Structures 2010* (fib, 2010),
- CEB-FIP Model Code 1990 (CEB, 1990), or
- ACI 209.2R-08.

For segmentally constructed bridges, estimates of shrinkage and creep may be made using the provisions of the *fib Model Code for Concrete Structures 2010* (fib, 2010), or the CEB-FIP Model Code 1990 (CEB, 1990). In addition, a more precise estimate shall be made, including the effect of all of the following:

- specific materials, when known,
- structural dimensions,
- site conditions,
- construction methods, and
- concrete age at various stages of erection.

C5.4.2.2

The thermal coefficient depends primarily on the types and proportions of aggregates used and on the degree of saturation of the concrete.

The thermal coefficient of normal weight concrete can vary between 3.0 to $8.0 \times 10^{-6}/^{\circ}\text{F}$, with limestone and marble aggregates producing the lower values, and chert and quartzite the higher. Only limited determinations of these coefficients have been made for lightweight concretes. They are in the range of 4.0 to $6.0 \times 10^{-6}/^{\circ}\text{F}$ and depend on the amount of natural sand and type of lightweight aggregate used.

Additional information may be found in ACI 343 (1995), ACI 209 (2008), and ACI 213 (2014).

C5.4.2.3.1

Creep and shrinkage of concrete are variable properties that depend on a number of factors, some of which may not be known at the time of design.

Without specific physical tests or prior experience with the materials, the use of the empirical methods referenced in these specifications cannot be expected to yield results with errors less than ± 50 percent.

Values of modulus of elasticity, creep factors, and shrinkage factors should be taken from a consistent source.

Comparisons of creep coefficients show that Article 5.4.2.3.2 and ACI 209 estimate lower creep coefficients than the *fib Model Code for Concrete Structures 2010* (fib, 2010) and the CEB-FIP Model Code 1990 (CEB, 1990) in segmentally constructed bridges. The use of the *fib Model Code for Concrete Structures 2010* (fib, 2010) is recommended by the American Segmental Bridge Institute for creep and shrinkage calculations of segmentally constructed bridges.

Where mix-specific data are available, more precise models can be utilized for creep and shrinkage estimates for segmental bridges, including the B4 model from Wendner et al. (2013).